NATIONAL INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
END SEMESTER EXAMINATION (ACADEMIC YEAR 2025-2026)

Course: CS-301 Operating Systems | Max Marks: 50 | Duration: 3 Hours

SECTION A (Short Answer Questions - Answer all)

1. Define a deadlock and state the four necessary Coffman conditions. [2 Marks]

2. What is the primary role of the Translation Lookaside Buffer (TLB) in paging? [2 Marks]

3. State the difference between preemptive and non-preemptive CPU scheduling algorithms. [2 Marks]

4. Define an inode in UNIX file systems and state what information it stores. [2 Marks]

SECTION B (Analytical & Application Questions)

5. Explain the step-by-step execution of Banker's Algorithm for deadlock avoidance with 3 processes. [6 Marks]

6. Calculate the average turnaround time and waiting time for Round Robin scheduling with time quantum q=4. [6 Marks]

7. Apply the FIFO and Optimal page replacement algorithms to the given reference string for 3 page frames. [6 Marks]

SECTION C (Comprehensive Design & System Evaluation)

8. Analyze the producer-consumer problem using POSIX semaphores and identify where race conditions can arise. [10 Marks]

9. Architect a reader-writer synchronization locking primitive in C that prevents writer starvation. [12 Marks]

10. Critique the security and performance trade-offs of microkernel architectures versus monolithic kernels. [12 Marks]